

Claude Desktop (MCP Client)



Orchestra — LangGraph MCP Server

MCP Server to Claude Desktop · MCP Client to RegNetAgents & CASCADE · 3 Composite Tools

Decision Layer

Classify Gene
(CASCADE metadata)

Route Analysis
(gene role)

TF path
effector path
validation path

Evidence Layer — Parallel MCP Calls
RegNetAgents

comprehensive_gene_analysis(gene, cell_type)
pathway_focused_analysis · query_network(gene, cell_type)

CASCADE

comprehensive_perturbation_analysis(gene, cell_type)
get_gene_metadata · get_protein_interactions · therapeutic_target_discovery

Explanation Layer

Synthesize
cross-system corroboration count

Generate Report
structured evidence table + optional LLM narrative

RegNetAgents MCP Server
network analysis · pathway enrichment
PageRank centrality · domain agents
(Bird et al. 2026a)

CASCADE MCP Server
perturbation simulation · STRING PPI
LINCS L1000 · DepMap CRISPR · super-enhancers
DoRothEA · cBioPortal (Bird et al. 2026b)